

Math 675 Spring 2026:
Homework 4 (LATE)
Due Tuesday, April 6th 2026

Question 1:

Consider the matrix $A = \begin{pmatrix} -5 & 1 \\ 4 & -2 \end{pmatrix}$

The eigenvalues for A are $\lambda_1 = -1$ and $\lambda_2 = -6$ with corresponding eigenvectors $\mathbf{v}_{\lambda_1} = (1, 4)$ and $\mathbf{v}_{\lambda_2} = (-1, 1)$. Use this and the method of undetermined coefficients to find the general solution to the inhomogeneous system:

$$\mathbf{x}'(t) = \begin{pmatrix} -5 & 1 \\ 4 & -2 \end{pmatrix} \mathbf{x}(t) + e^{2t} \begin{pmatrix} 6 \\ -1 \end{pmatrix}$$

Question 2:

Consider the matrix $A = \begin{pmatrix} 2 & -1 \\ 3 & 2 \end{pmatrix}$

The eigenvalues for A are $\lambda_1 = -1$ and $\lambda_2 = 1$ with corresponding eigenvectors $\mathbf{v}_{\lambda_1} = (1, 3)$ and $\mathbf{v}_{\lambda_2} = (1, 1)$.

1. Calculate $\exp(tA)$

Given the definition: $\exp(tA) = \Psi(t)\Psi^{-1}(0)$, where $\Psi(t)$ is the fundamental matrix made up of the x and y solution vectors from $\mathbf{x}(t)$.

First, build the general solution:

$$\mathbf{x}(t) = c_1 e^{-t} \begin{pmatrix} 1 \\ 3 \end{pmatrix} + c_2 e^t \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$\Psi(t) = \begin{pmatrix} e^{-t} & e^t \\ 3e^{-t} & e^t \end{pmatrix}$$

$$\Psi^{-1}(0) = (\Psi(0))^{-1} = \begin{pmatrix} 1 & 1 \\ 3 & 1 \end{pmatrix}^{-1} = \frac{1}{2} \begin{pmatrix} -1 & 1 \\ 3 & -1 \end{pmatrix}$$

$$\exp(tA) = \Psi(t)\Psi^{-1}(0) = \begin{pmatrix} e^{-t} & e^t \\ 3e^{-t} & e^t \end{pmatrix} \frac{1}{2} \begin{pmatrix} -1 & 1 \\ 3 & -1 \end{pmatrix}$$

$$\exp(tA) = \frac{1}{2} \begin{pmatrix} -e^{-t} + 3e^t & e^{-t} - e^t \\ -3e^{-t} + 3e^t & 3e^{-t} - e^t \end{pmatrix}$$

Now, optionally, we could convert this into a form involving hyperbolic trig:

$$\exp(tA) = \begin{pmatrix} \sinh(t) & -\sinh(t) \\ 3\sinh(t) & \cosh(t) - 2\sinh(t) \end{pmatrix}$$

2. Solve the IVP

Use the result from part (1) to solve:

$$\mathbf{x}'(t) = \begin{pmatrix} 2 & -1 \\ 3 & -2 \end{pmatrix} \mathbf{x}(t), \quad \mathbf{x}(0) = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

Defintion: Given a system $\mathbf{x}'(t) = A\mathbf{x}(t)$ the solution is $\mathbf{x}(t) = e^{tA}\mathbf{x}(0)$

$$\mathbf{x}(t) = e^{tA}\mathbf{x}(0) = \frac{1}{2} \begin{pmatrix} -e^{-t} + 3e^t & e^{-t} - e^t \\ -3e^{-t} + 3e^t & 3e^{-t} - e^t \end{pmatrix} \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

$$\mathbf{x}(t) = e^{tA}\mathbf{x}(0) = \frac{1}{2} \begin{pmatrix} -e^{-t} + 3e^t + 2e^{-t} - 2e^t \\ -3e^{-t} + 3e^t + 6e^{-t} - 2e^t \end{pmatrix}$$

$$\mathbf{x}(t) = e^{tA}\mathbf{x}(0) = \frac{1}{2} \begin{pmatrix} e^{-t} + e^t \\ 3e^{-t} + e^t \end{pmatrix}$$

$$\mathbf{x}(t) = \frac{1}{2}e^{-t} \begin{pmatrix} 1 \\ 3 \end{pmatrix} + \frac{1}{2}e^t \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

Ta-Da! This is the same IVP Solution as if we solved this the regular way!

Question 3:

The following questions refer to the proof of existence and uniqueness covered in chapter 2.3 of the guided notes.

a) Why do we choose a small time interval?

In the proof we chose $\alpha < \min \left\{ \frac{\rho}{M}, \frac{1}{B} \right\}$

(a) Explain in words why we need the condition $\alpha < \frac{\rho}{M}$

(b) Explain in words why we need the condition $\alpha < \frac{1}{B}$

(c) What could go wrong if α was too large?

b) Staying inside the closed ball

Suppose we have $x_{k+1}(t) = x_0 + \int_0^t F(x_k(x))ds$

(a) Why is it important that all iterates stay inside $\mathcal{O}_\rho$

- (b) What assumptions about F would fail if an iterate left $\mathcal{O}_\rho$?
- (c) Explain how the bound $\|F(w)\| \leq M$ is used in this step
- c) **Role of Lipschitz Condition**
The proof uses $\|F(x) - F(y)\| \leq B\|x - y\|$
- (a) Where in the proof is this condition used?
- (b) Why is continuity of F alone not enough for the argument?
- (c) In your own words, what does the Lipschitz condition guarantee?
- d) **Why does a geometric series appear?**
We obtained an estimate $\|x_{k+1}(t) - x_k(t)\| \leq (\alpha B)^k L$
- (a) Why is it important that $\alpha B < 1$?
- (b) What would happen if $\alpha B \geq 1$?
- (c) How does this estimate lead to convergence of $\{x_k\}$?
- e) **Conceptual comparison**
Compare:
- Solving an algebraic equation $x = G(x)$ via fixed point iteration
 - Solving the integral equation $x(t) = x_0 + \int_0^t F(x(s))ds$
- (a) What plays the role of the function being iterated?
- (b) What plays the role of the fixed point?
- (c) Why is this analogy useful?